Civics Group	Index Number	Name (use BLOCK LETTERS)

H2

ST. ANDREW'S JUNIOR COLLEGE
2017 JC2 BT2

H2 BIOLOGY **9744/2**

Paper 2

Tuesday 12th September 2017 2 hours

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagram, graph or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A (Structured Questions)
Answer **all** questions.
Write your answers in the spaces provided on the question paper.

The number of marks is given in brackets [ ] at the end of each question or part question.

Conceptual error (CE)	Lack of Keywords (K)	Misreading the question (Q)

For Examiners' Use	
Section A	
1	/18
2	/14
3	/12
4	/12
5	/12
6	/12
7	/10
8	/10
Total	**/100**

This document consists of **24** printed pages.

[Turn over

QUESTION 1

Hormones, insulin and glucagon, are proteins that regulate the concentration of blood glucose level. Type 2 diabetes is characterized both by insulin resistance, a condition in which various tissues in the body no longer respond properly to insulin action, and by subsequent progressive decline in beta (β)-cell function to the point that the cells can no longer produce enough additional insulin to overcome the insulin resistance. Researchers are actively exploring use of stem cells as a potential source of deriving new β-cells to treat type 2 diabetes.

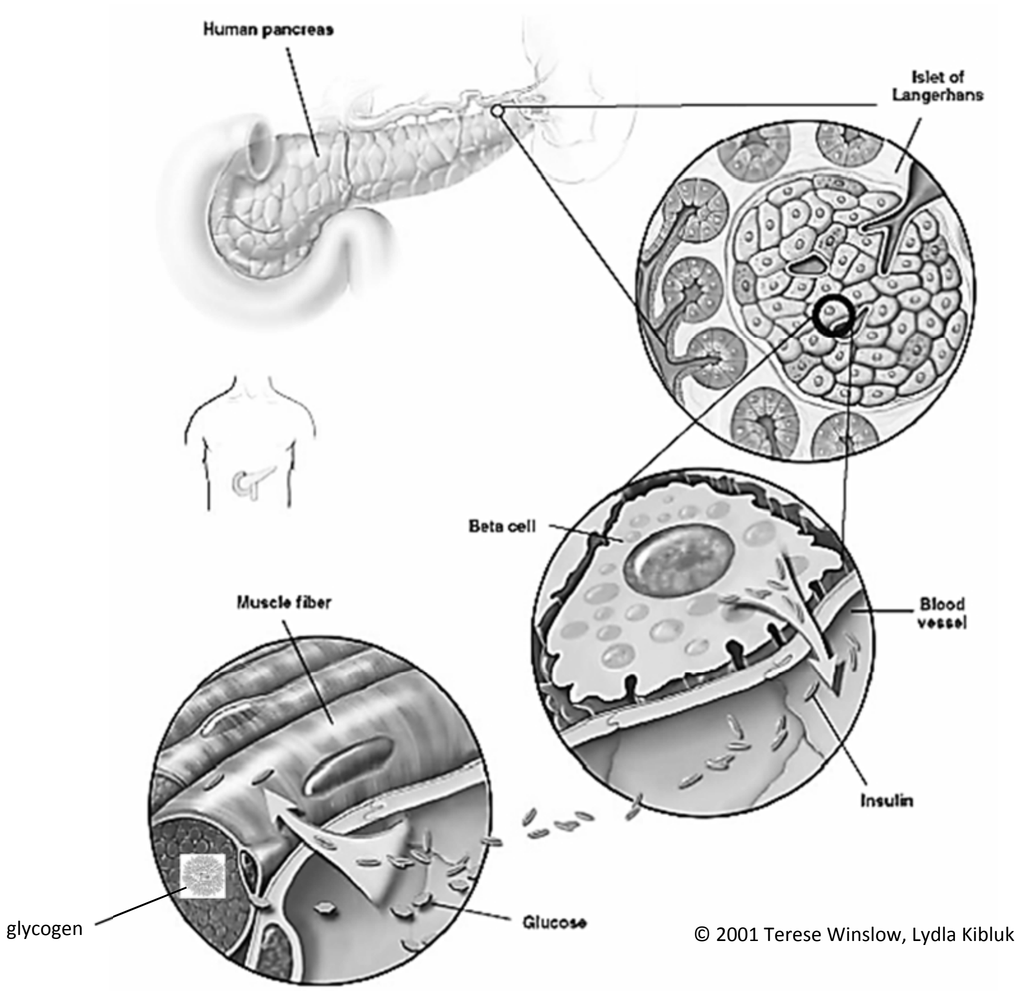

Fig. 1

The pancreas is located in the abdomen, adjacent to the duodenum (the first portion of the small intestine). A cross section of the pancreas shows the islet of Langerhans which is the functional unit of the endocrine pancreas. Encircled is the beta cell that synthesizes and secretes insulin. Beta cells are located adjacent to blood vessels and can easily respond to changes in blood glucose concentration by adjusting insulin production.

(a) Cells that secrete proteins contain a lot of rough endoplasmic reticulum (rER) and a large Golgi body.

(i) Describe how the rER is involved in the production of insulin.

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(ii) Describe how the Golgi body is involved in the secretion of insulin.

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(b) Using type II diabetes as an example, explain how environment affects phenotype.

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(c) With reference to Fig. 1, explain how the binding of insulin to receptors on muscle cells leads to a lowering of blood glucose concentration.

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(d) Suggest how a change in the amino acid sequence of the receptor found in the plasma membrane of the muscle cell could make the cell resistant to insulin.

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(e) Describe how phospholipids are arranged in a plasma membrane.

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(f) Phospholipids are a type of lipid. Lipids, in general, are made up of glycerol and fatty acids monomers covalently bonded together. Name the covalent bond and describe the breakage of this bond.

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Experiments have indicated that pancreatic stem cells (PSCs) can serve as sources of insulin secreting cells.

(g) State the source of PSCs and explain the PSCs' normal functions.

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(h) Suggest an advantage of using the patient's own PSCs to regenerate tissue or organs.

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[Q1: 18 marks]

QUESTION 2

Stephen Lyon Crohn, also known as "The man who can't catch AIDS", was a man notable for a genetic mutation, which caused his inability to contract AIDS.

Crohn had the "delta-32" mutation on the CCR5 co-receptor, a protein on the surface of cells involved in the immune system.

CCR5 “delta-32” mutation involves a 32-base-pair deletion in the CCR5 co-receptor gene locus.

(a) Describe the role of CCR5 co-receptor protein in the entry of Human HIV into host cells.

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(b) Describe briefly how the information on DNA is used to synthesize the pre-mRNA transcript for CCR5 co-receptor.

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Individuals can either gain full or partial resistance to HIV, depending on the number of mutated CCR5 alleles they possess at the CCR5 gene locus. Partial resistance to HIV occurs when the entry of the virus is severely hampered (but not completely prohibited).

Another Caucasian woman named Susan, is homozygous with the CCR5 “delta-32” mutation. Her husband has the same mutation on **one** CCR5 allele. The couple has a son and a daughter. Both son and daughter has 50% chance of having partial resistance and 50% chance of having full resistance.

(c) Explain if the inheritance of the CCR5 alleles is autosomal or X-linked.

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(d) Suggest why some individuals have partial resistance to HIV.

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The son with **partial resistance** to HIV, marries a woman with **no resistance**.

The son has blood group O while his wife is heterozygous for blood group A.

(e) With the help of a genetic diagram, show all the possible outcomes for their offspring. [5]

[Q2: 14 marks]

QUESTION 3

Epidermal growth factor (EGF) is released by cells, and is picked up either by the cell itself or by neighbouring cells. It regulates the production of a number of proteins in target cells. Protein produced and its effect depends on the type of target cell.

Fig. 3 shows how EGF regulates 3 genes.

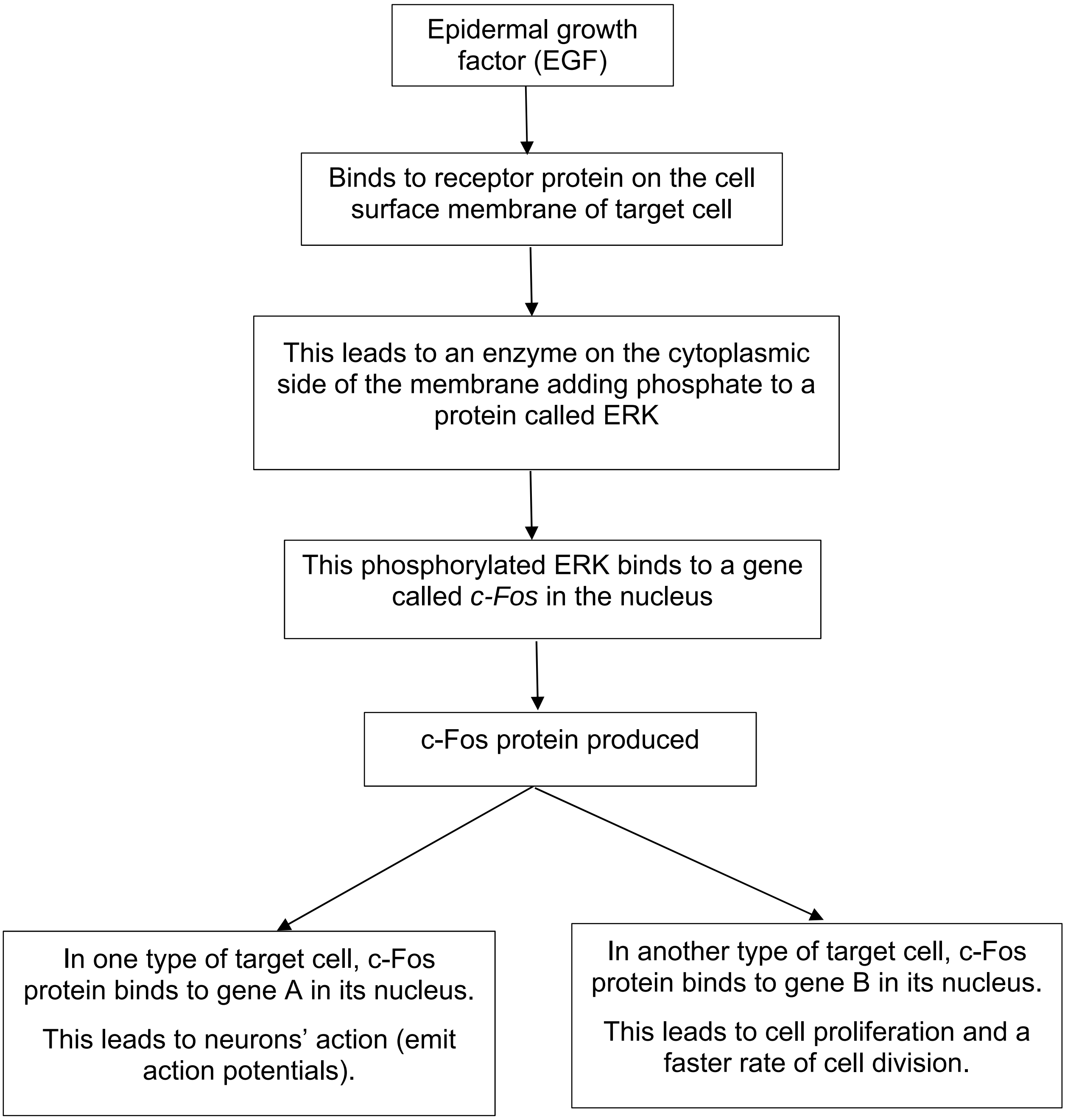

Fig. 3

(a) Name the **two** transcription factors in Fig. 3.

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(b) The *c-Fos* gene can be a proto-oncogene.

Use the information in Fig. 3 to explain how a mutation of the *c-Fos* gene can result in the formation of a tumour.

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(c) Gene B has been associated with a significant number of human cancers. Scientists used polymerase chain reaction (PCR) to make multiple copies of gene B extracted from a patient's cancer tissue sample.

The reaction mixture includes the sample of DNA to be copied plus the following ingredients:

- DNA primers
- buffer solution
- heat-stable DNA polymerase (Taq polymerase)
- deoxyribonucleoside triphosphates (deoxyATP, deoxyTTP, deoxyCTP and deoxyGTP)

(i) Suggest why a buffer needs to be present in the reaction mixture.

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(ii) The deoxyribonucleoside triphosphates that are added to the reaction mixture are the monomers used for making the new DNA strands.

Suggest **one further** reason for adding the deoxyribonucleoside triphosphates to the reaction mixture.

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(iii) In the first stage of PCR, the mixture is heated to a temperature of around 90ºC to denature the DNA. Suggest why high temperatures are needed to separate the two DNA strands.

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(iv) At the end of several cycles of PCR, many copies of the DNA sample in the reaction mixture will have been made. The DNA samples are then separated out to produce a DNA banding pattern.

State the technique used to separate out the DNA samples **and** describe how this technique works.

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[Q3: 12 marks]

QUESTION 4

Researchers have identified a gene that gives bacteria resistance to a type of antibiotics called polymyxins. Despite being discovered around 60 years ago, polymyxins maintained their effectiveness as antibiotics as they were seldom used due to concerns about their toxicity.

In recent years, rampant use of common antibiotics (e.g. penicillin) has led to the emergence of bacterial strains which are resistant to these antibiotics. This has become more and more of a global concern. Polymyxins are now a last line of defense against bacteria because of its previous lack of use.

(a) With reference to the reproductive cycle of bacteriophages, suggest how bacteriophage infections may lead to a spread of antibiotic resistance between bacterial populations.

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Bacteria reproduce by the process of binary fission.

(b) Explain the significance of binary fission in bacteria.

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The process of binary fission involves semi-conservative DNA replication.

(c) State two differences in the formation of the leading and lagging strands during DNA replication.

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Bacteria rely on sugar sources e.g. lactose for survival.

(d) Describe the consequence of mutating the *lacI* gene of the bacterial lac operon, on usage of lactose.

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[Q4: 12 marks]

QUESTION 5

A student set up an experiment to investigate the effect of carbon dioxide on photosynthesis. First, he de-starched a small potted plant by putting it in the dark for two days. Then, he chose two leaves and inserted them into conical flasks, **A** and **B**, fitted with rubber stoppers. Lithium hydroxide was placed in Flask **B** to absorb all carbon dioxide present. The plant was then left under a table lamp for 15 minutes. Fig 5.1 shows the experimental setup.

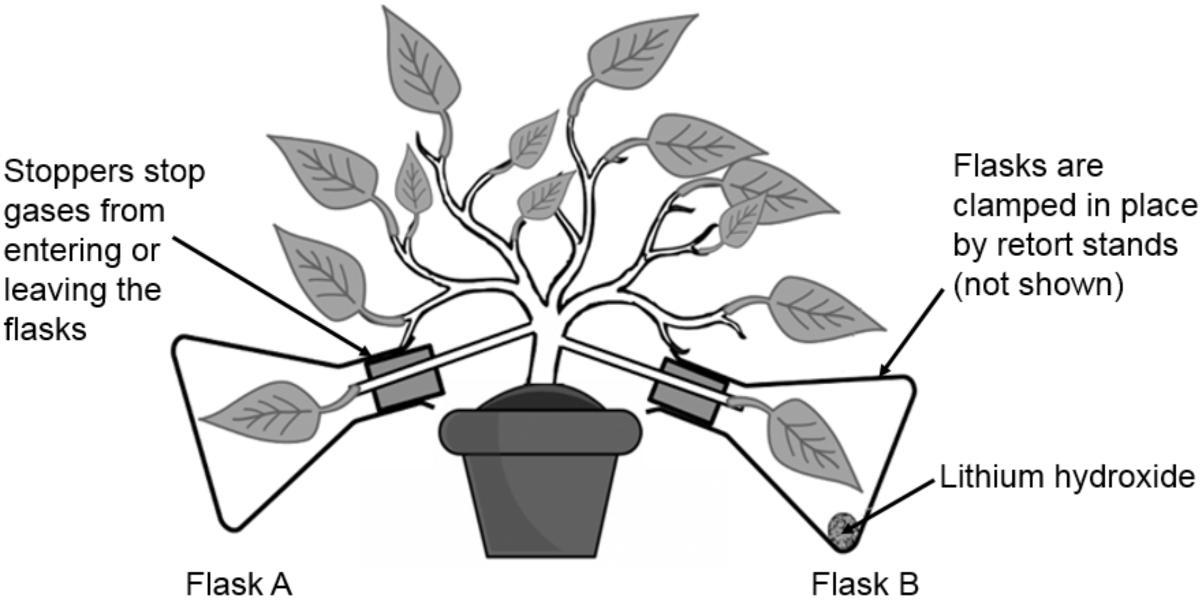

Fig 5.1

He removed a sample of each leaf every 5 minutes (by punching out a leaf disc of approximately 0.2 cm in diameter, using a single-hole puncher) and return the leaves to their respective flasks immediately. Each leaf disc was then tested for the presence of ribulose bisphosphate (RuBP) and starch. Table 5.2 shows the results he obtained.

Table 5.2

Flask	Concentration of RuBP / μ molm^{-2}				Concentration of starch / μ molm^{-2}			
	0 min	5 min	10 min	15 min	0 min	5 min	10 min	15 min
A	0.0	2.2	3.0	3.1	0.0	2.1	3.4	6.5
B	0.0	2.7	4.2	6.8	0.0	0.3	0.5	0.5

(a) State two other variables which must be kept constant to maximize the validity of the results obtained for this experiment.

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(b) With reference to Table 5.2, describe the relationship between the presence of carbon dioxide and concentration of starch.

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(c) Explain the absence of RuBP in both leaves at the start of the experiment.

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(d) The increase in RuBP concentration for the leaf in Flask A reached a plateau from 10 min to 15 min of exposure to light but continued to increase in the leaf in Flask B up to 15 min. Explain why.

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The student watered the potted plant too excessively, causing the soil to become waterlogged. Fortunately, the roots of this plant could carry out anaerobic respiration under low oxygen conditions in the soil.

(e) Outline the process of anaerobic respiration in the roots under waterlogged conditions.

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[Q5: 12 marks]

QUESTION 6

The maggot fly, *Rhagoletis pomonella*, is native to the United States, and up until the discovery of the Americas by Europeans, fed solely on hawthorns. But when Europeans arrived on the Americas, they brought with them a new potential food source for the flies: apples.

At first, the flies are not attracted to apples. But over time, a minuscule change in the connections of two channels in the brain - one for detecting hawthorn odours and the other for apple odours - caused some flies to switch host plant (i.e. they jumped to apple trees). It was observed that the maggot flies strongly preferred to mate and lay their eggs on the type of tree they were born on.

When geneticists took a closer look in the late 20th century, they found that the two types - those that feed on apples and those that feed on hawthorns - have become genetically distinct groups.

(a) Identify the type of reproductive isolation shown in this case study.

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(b) Discuss the processes that could have led to the genetic distinctions between the apple-eating flies and hawthorn-eating flies.

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(c) Suggest how researchers can conclusively determine whether the apple-eating flies are a different species from the hawthorn-eating flies.

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A nuclear small subunit (18S) rRNA gene has been sequenced for various insect species and used to reconstruct their evolutionary relationship, which is shown in a phylogenetic tree in Fig. 6.1.

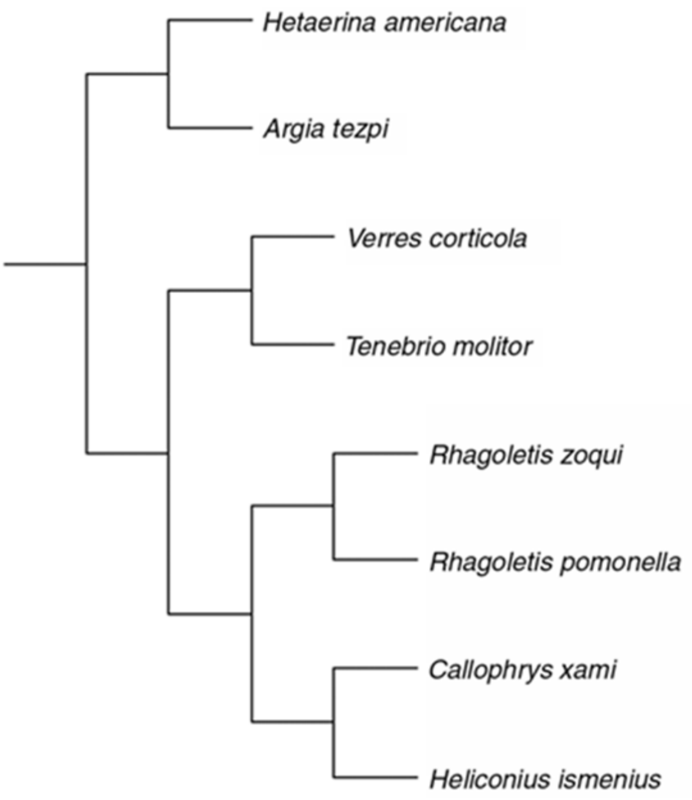

Fig. 6.1

(d) Comment on the evolutionary relationship between *Rhagoletis pomonella*, *Rhagoletis zoqui* and *Callophrys xami.*

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(e) Discuss two advantages of using 18S rRNA gene over morphological comparisons for the reconstruction of the insect phylogeny.

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[Q6: 12 marks]

QUESTION 7

Influenza is a prevalent illness all over the world. Due to the high rate of change in the structure of the virus, annual vaccinations are recommended.

To overcome this **problem** of the constant need for developing new vaccines against influenza viruses, scientists are attempting to produce vaccines relating to antibodies which recognize part of the virus that does not change—the stalk of haemagglutinin. This research may lead to the development of "universal" influenza vaccines which can remain effective over long periods of time (Journal of Virology, 2017).

Fig. 7.1 shows the structure of haemagglutinin.

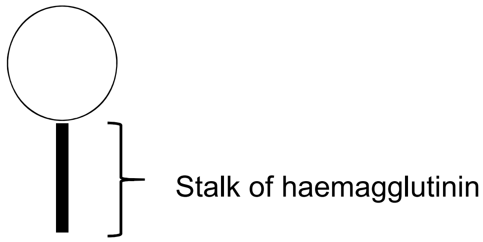

Fig. 7.1

(a) With reference to the context above, explain why there is a "constant need for developing new vaccines against influenza viruses".

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(b) With reference to the influenza virus and active immunity, define a vaccine and state how it provides protection against the virus.

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Vaccinations can control diseases by resulting in *herd immunity*, in which large percentage of population which has become immune to the virus through vaccinations, provides a measure of protection for individuals who are not immune.

(c) Explain why.

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Tuberculosis (TB) is an infectious disease that generally affects the lungs. Most infections are latent and do not have symptoms. Latent infections can progress to active form of the disease which, if left untreated, kills about half of those infected.

The most common classic symptom of active TB is chronic cough with blood-containing sputum.

(d) Name the organism which causes tuberculosis (TB).

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In the alveoli tissues, the bacteria that causes TB binds directly with mannose receptors on alveolar macrophages using a bacterial glycolipid, and is engulfed by alveolar macrophages.

(e) With reference to a cellular organelle in the macrophages, describe how macrophages attempt to process engulfed bacteria.

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(f) Describe the cause of the chronic cough symptom of TB in its **active** stage.

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[Q7: 10 marks]

QUESTION 8

Proteaceae are a family of flowering plants predominantly distributed in the tropical and sub-tropical regions of Australia and South Africa. Threatened species may be vulnerable to climate change, because of their narrow temperature tolerance, small population sizes and restricted distributions.

A study modelled climate-induced changes on the range size (geographical distribution of a species) of 282 Proteaceae species. Figures 8.1a and 8.1b show the time course of range changes for wind-dispersed (**A**) and ant-dispersed (**B**) Proteaceae species (under 'full migration' dispersal assumptions, which assumes no limitation to migration). Error bars represent standard error (for example, smaller standard errors indicates that the observations are closer to the actual value).

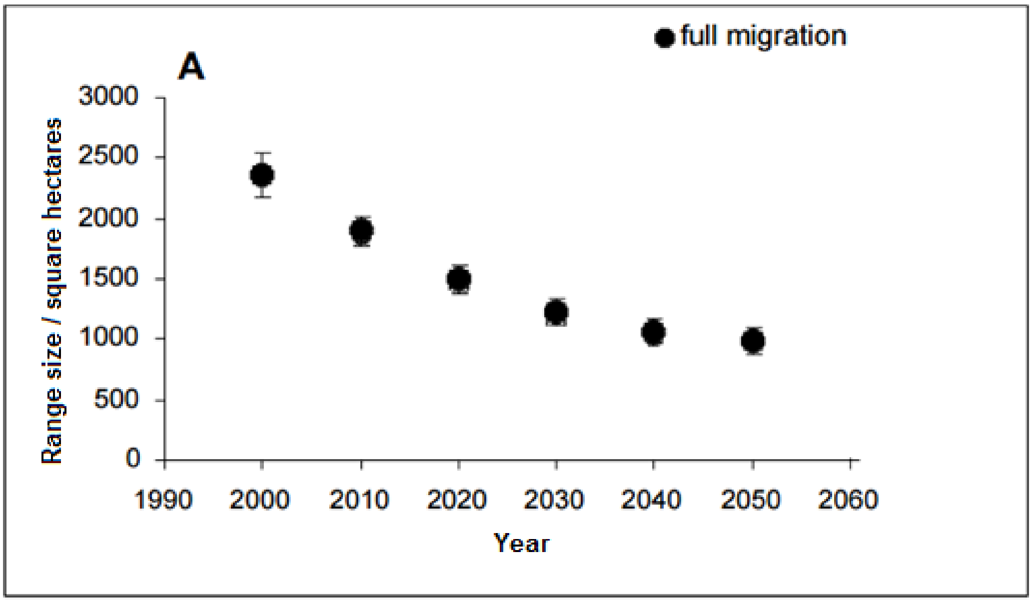

Fig. 8.1a

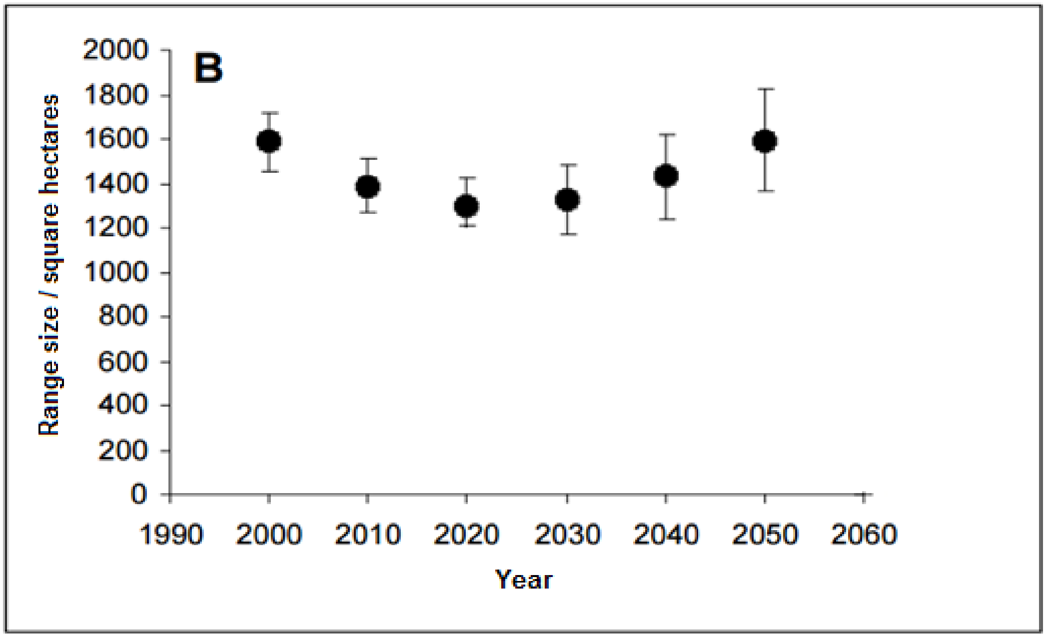

Fig. 8.1b

(a) Describe the differences in climate-induced changes in range size between the wind-dispersed and ant-dispersed Proteaceae species from Year 2000 to 2050.

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(b) The Intergovernmental Panel for Climate Change (IPCC) predicted an increase of 2°C in global temperatures by 2050. Suggest reasons for the larger range size of ant-dispersed Proteaceae species compared to that of wind-dispersed species in 2050.

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It was predicted that not only will the range size for ant-dispersed Proteaceae species remain high in 2050, it is likely that the plants would colonise new regions, particularly in the higher latitudes and altitudes.

(c) Explain why the Proteaceae species can potentially threaten the native species of the regions they expand into.

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Many indigenous cultures in tropical regions have used Proteaceae for medicinal preparations. For example, bioactive ingredients can be obtained from infusions of the roots, bark, leaves, or flowers of many Proteaceae species, which can used as topical applications for skin conditions or internally as tonics, aphrodisiacs, and medicines to treat headaches, cough, diarrhea, indigestion, stomach ulcers, and kidney disease.

(d) Discuss how climate change can affect the biodiversity of the Proteaceae species and its consequence to the production of biomedicines.

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[Q8: 10 marks]

END OF PAPER